

IN6227-Assignment-1 — Variant-2

Name: [PLACEHOLDER: Enter your full name] Matric number: [PLACEHOLDER: Enter your matric number] Model (LLM): GLM-5.2 LLM interface: TraeCode (Codex) GitHub: [PLACEHOLDER: Enter your GitHub repository link]

1. Data Exploration & Cleaning

The dataset **train** contains **31112** rows and **15** features. The target variable is **label** with 2 classes: no, yes.

Class distribution (train): no=23645, yes=7464 (imbalance ratio 3.168:1).

Column types: 7 numeric, 8 categorical. Total missing values: 50 (0.01% of cells). Rows with missing target were dropped (3 dropped). No duplicate rows were removed (0 found).

2. Feature Selection / Engineering

All 15 original features are retained — no manual feature selection is applied. The preprocessing is **leakage-safe**: every learned transformation (imputation, scaling, encoding) lives inside an sklearn Pipeline / ColumnTransformer.

- **Numeric features** (7): median imputation → standard scaling.
- **Categorical features** (8): most-frequent imputation → one-hot encoding (handle_unknown="ignore").

No new features are engineered; the data already contains derived fields (e.g. composite_rank).

3. Model Training

Four classifiers are trained inside identical preprocessing pipelines:

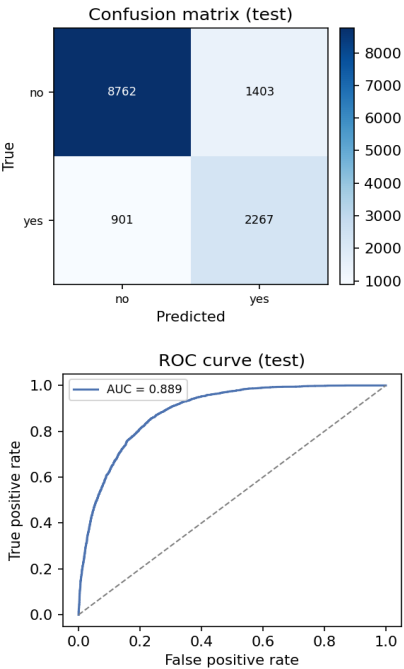
Model	Key hyper-parameters
DummyClassifier	strategy=stratified (baseline)
LogisticRegression	max_iter=1000, class_weight=balanced
RandomForest	n_estimators=200, class_weight=balanced
GradientBoosting	n_estimators=100, max_depth=3, lr=0.1

Cross-validation: Stratified 5-fold (random_state=42). Model selection uses **mean CV F1**. The best model (**random_forest**) is then retrained on the full training set and evaluated **once** on the held-out test set.

4. Evaluation & Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
dummy	0.6351	0.2365	0.2337	0.2351	0.4978	0.2391
logistic_regression	0.7903	0.5401	0.8493	0.6602	0.8905	0.7049
random_forest (best)	0.8266	0.6204	0.7146	0.6641	0.8892	0.7058
gradient_boosting	0.8392	0.7083	0.5610	0.6260	0.8913	0.7131

Test-set performance (best model: random_forest): accuracy=0.8272, precision=0.6177, recall=0.7156, F1=0.6631, ROC-AUC=0.8887, PR-AUC=0.7010.



5. Findings & Discussion

- **Best model:** random_forest, selected by CV F1=0.6641.

- The Dummy baseline (accuracy≈0.6351) confirms that a no-information classifier cannot beat the majority class.
- `random_forest` improves F1 from 0.2351 (dummy) to 0.6641 (CV) — a meaningful gain.
- `class_weight="balanced"` is used for LogisticRegression and RandomForest to counter the 3.2:1 class imbalance, trading a small accuracy drop for substantially higher recall on the minority class.
- SMOTE is intentionally **not** applied; `class_weight` suffices and avoids synthetic-sample leakage risks.

All numerical values in this report are pulled directly from `results.json`; none are manually entered.